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MANIFOLD TO SELECTIVELY INFLATE AIRBAGS

Abstract

A vehicle includes a first seat and a first airbag supported by the first seat. The vehicle includes a second seat and a second airbag supported by the second seat. The vehicle includes a middle console disposed between the first seat and the second seat. The vehicle includes a manifold supported by the middle console. The manifold defines a chamber and including a first fluid outlet operatively coupled to the first airbag to provide fluid from the chamber to the first airbag and a second fluid outlet operatively coupled to the second airbag to provide fluid from the chamber to the second airbag. The vehicle includes an inflator supported by the middle console, the inflator operatively coupled to the manifold to provide fluid to the chamber.

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Background/Summary

BACKGROUND

[0001] A vehicle may include one or more airbags deployable during certain vehicle impacts to control kinematics of occupants inside the vehicle during the impact. The airbag may be a component of an airbag assembly including a housing supporting the airbag, and an inflator in communication with the airbag for inflating the airbag from an uninflated position to an inflated position.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0002] FIG. 1 is a top view of a portion of a vehicle broken away to show a passenger cabin with a manifold and airbags at uninflated positions.

[0003] FIG. 2 is a top view of the portion of the vehicle broken away to show the passenger cabin with the airbags at inflated positions.

[0004] FIG. 3 is a diagrammatic view of the manifold with valves at closed positions.

[0005] FIG. 4 is a diagrammatic view of the manifold with the valves at open positions.

[0006] FIG. 5 is a block diagram of components of the vehicle.

[0007] FIG. 6 is a top view of the portion of the vehicle broken away to show the passenger cabin with certain of the airbags at inflated positions.

[0008] FIG. 7 is a top view of the portion of the vehicle broken away to show the passenger cabin with other certain of the airbags at inflated positions.

[0009] FIG. 8 is a top view of the portion of the vehicle broken away to show the passenger cabin with yet other certain of the airbags at inflated positions.

DETAILED DESCRIPTION

[0010] A vehicle includes a first seat and a first airbag supported by the first seat. The vehicle includes a second seat and a second airbag supported by the second seat. The vehicle includes a middle console disposed between the first seat and the second seat. The vehicle includes a manifold supported by the middle console. The manifold defines a chamber. The manifold includes a first fluid outlet. The first fluid outlet is operatively coupled to the first airbag to provide fluid from the chamber to the first airbag. The manifold includes a second fluid outlet. The second fluid outlet is operatively coupled to the second airbag to provide fluid from the chamber to the second airbag. The vehicle includes an inflator supported by the middle console. The inflator is operatively coupled to the manifold to provide fluid to the chamber.

[0011] The manifold may include an exhaust fluid outlet.

[0012] The vehicle may include an exhaust valve movable to an open position permitting fluid flow through the exhaust fluid outlet and to a closed position inhibiting fluid flow through the exhaust fluid outlet.

[0013] The vehicle may include a first valve movable to an open position permitting fluid flow through the first fluid outlet and to a closed position inhibiting fluid flow through the first fluid outlet, and a second valve movable to an open position permitting fluid flow through the second fluid outlet and to a closed position inhibiting fluid flow through the second fluid outlet.

[0014] The vehicle may include a computer having a processor and memory storing instructions executable by the processor to, in response to detecting a certain impact to the vehicle, command at least one of the first valve or the second valve from the closed position to the open position and command actuation of the inflator.

[0015] The instructions may include instructions to, after a threshold amount of time has lapsed

from commanding the at least one of the first valve or the second valve from the closed position to the open position, command the at least one of the first valve or the second valve from the open positions to the closed positions.

[0016] The threshold amount of time may be 20 to 60 milliseconds.

[0017] The instructions may include instructions to command the exhaust valve from the closed position to the open position after a second threshold amount of time has lapsed from detecting the certain impact.

[0018] The second threshold amount of time may be at least two seconds.

[0019] The vehicle may include a third airbag supported by the middle console and a fourth airbag supported by the middle console, and the manifold may include a third fluid outlet operatively coupled to the third airbag to provide fluid from the chamber to the third airbag and a fourth fluid outlet operatively coupled to the fourth airbag to provide fluid from the chamber to the fourth airbag.

[0020] The first seat and the second seat may each include a seatback, the first airbag and the second airbag may be inflatable to inflated positions that extend forward from the seatbacks of the first seat and the second seat, and the third airbag and the fourth airbag may be inflatable to inflated positions that extend laterally outboard from the middle console in front of the first seat and the second seat.

[0021] The manifold may include an exhaust fluid outlet.

[0022] The vehicle may include an exhaust valve moveable to an open position permitting fluid flow through the exhaust fluid outlet and to a closed position inhibiting fluid flow through the exhaust fluid outlet.

[0023] The vehicle may include a first valve movable to an open position permitting fluid flow through the first fluid outlet and to a closed position inhibiting fluid flow through the first fluid outlet, a second valve movable to an open position permitting fluid flow through the second fluid outlet and to a closed position inhibiting fluid flow through the second fluid outlet, a third valve movable to an open position permitting fluid flow through the third fluid outlet and to a closed position inhibiting fluid flow through the third fluid outlet, and a fourth valve movable to an open position permitting fluid flow through the fourth fluid outlet and to a closed position inhibiting fluid flow through the fourth fluid outlet.

[0024] The vehicle may include a computer having a processor and memory storing instructions executable by the processor to, in response to detecting a certain impact to the vehicle, command at least one of the first valve, the second valve, the third valve, or the fourth valve from a closed position to an open position and command actuation of the inflator.

[0025] The instructions may include instructions to, in response to detecting another certain impact to the vehicle, command another of the at least one of the first valve, the second valve, the third valve, or the fourth valve from the closed position to the open position.

[0026] The instructions may include instructions to command the exhaust valve from the closed position to the open position after a threshold amount of time has lapsed from detecting the certain impact.

[0027] The threshold amount of time may be at least two seconds.

[0028] The chamber may define a first cross-sectional area, and the first fluid outlet may define a second cross-sectional area that is less than the first cross-sectional area.

[0029] With reference to the Figures, wherein like numerals indicate like parts throughout the several views, a vehicle **10** is shown. The vehicle **10** includes a first seat **12** and a first airbag **14** supported by the first seat **12**. The vehicle **10** includes a second seat **16** and a second airbag **18** supported by the second seat **16**. The vehicle **10** includes a middle console **20** disposed between the first seat **12** and the second seat **16**. The vehicle **10** includes a manifold **22** supported by the middle console **20**. The manifold **22** defines a chamber **24** and includes a first fluid outlet **26** operatively coupled to the first airbag **14** to provide fluid from the chamber **24** to the first airbag **14** and a

second fluid outlet **28** operatively coupled to the second airbag **18** to provide fluid from the chamber **24** to the second airbag **18**. The vehicle **10** includes an inflator **30** supported by the middle console **20**. The inflator **30** is operatively coupled to the manifold **22** to provide fluid, such as inflation medium, to the chamber **24**.

[0030] The manifold **22** may temporarily store inflation medium from the inflator **30** to provide increased control of inflation dynamics of the first airbag **14** and the second airbag **18**, e.g., control of inflation timing and inflation pressure of the airbags **14**, **18**. The manifold **22** may store the inflation medium for a sufficient amount of time to enable inflation of, e.g., the first airbag **14** in response to a certain impact to the vehicle **10**, and then after lapse of time, inflation of the second airbag **18** in response to detecting a certain subsequent impact to the vehicle **10**.

[0031] With reference to FIGS. **1** and **2**, the vehicle **10** may be any suitable type of automobile, e.g., a passenger or commercial automobile such as a sedan, a coupe, a truck, a sport utility vehicle, a crossover vehicle, a van, a minivan, a taxi, a bus, etc.

[0032] In the present description, relative vehicular orientations and directions (by way of example, top, bottom, front, rear, outboard, inboard, inward, outward, forward, rearward, lateral, left, right, etc.) are from the perspective of an occupant seated in the vehicle **10** and facing forward, e.g., toward a forward windshield of the vehicle **10**. The forward direction of the vehicle **10** is the direction of movement of the vehicle **10** when the vehicle **10** is engaged in forward drive with wheels of the vehicle **10** straight.

[0033] The vehicle **10** defines a lateral axis **A1** extending between a left-side and a right-side of the vehicle **10**. The vehicle **10** defines a longitudinal axis **A2** extending between a front and a rear of the vehicle **10**. The vehicle **10** defines a vertical axis extending between a top and a bottom of the vehicle **10**. The lateral axis **A1**, the longitudinal axis **A2**, and the vertical axis are perpendicular relative to each other.

[0034] The vehicle **10** includes a frame and a body. The body may be of unitary construction, in which the frame is unitary with the body, e.g., including frame rails, rockers, pillars, roof rails, etc. As another example, the body and frame may have a body-on-frame construction (also referred to as a cab-on-frame construction) in which the body (including rockers, pillars, roof rails, etc.) and frame are separate components, i.e., are modular, and the body is supported on and affixed to the frame. Alternatively, the frame and body may have any suitable construction. The frame and the body may be of any suitable material, for example, steel, aluminum, and/or fiber-reinforced plastic, etc. The body defines the passenger cabin to house occupants, if any, of the vehicle **10**. The passenger cabin may extend across the vehicle **10**, i.e., from one side to the other side of the vehicle **10**. The passenger cabin includes the front end and a rear end.

[0035] One or more seats **12**, **16**, e.g., the first seat **12** and the second seat **16**, may be supported in the passenger cabin, e.g., by the floor of the vehicle **10**. The first seat **12** may be at a first side, e.g., a driver side, of the vehicle **10**. The second seat **16** may be at a second side opposite the first side, e.g., a passenger side, of the vehicle **10**.

[0036] Each seat **12**, **16** includes a seatback **32** and a seat bottom **34** that can support an occupant of the respective seat **12**, **16**. For example, the occupant of the seat **12**, **16** may sit atop a top surface of the seat bottom **34** and recline against the seatback **32**. The seatback **32** and the seat bottom **34** may be adjustable in multiple degrees of freedom. Specifically, the seatback **32** and the seat bottom **34** may themselves be adjustable, in other words, adjustable components within the seatback **32** and/or the seat bottom **34**.

[0037] The seatback **32** can include a frame and a covering. The frame may include tubes, beams, etc. Specifically, the frame may include a pair of upright frame members. The upright frame members are elongated, and specifically, are elongated in a generally upright direction (e.g., generally along the vertical axis) when the seatback **32** is in a generally upright position. The upright frame members are spaced from each other along the lateral axis **A1**. The frame may include one or more cross-members extending between the upright frame members. The frame,

including the upright frame members, may be of any suitable plastic material, e.g., carbon fiber reinforced plastic (CFRP), glass fiber-reinforced semi-finished thermoplastic composite (organosheet), etc. As another example, some or all components of the frame may be formed of a suitable metal, e.g., steel, aluminum, etc. The cover may include upholstery and padding. The upholstery may be cloth, leather, faux leather, or any other suitable material. The upholstery may be stitched in panels around the frame. The padding may be between the cover and the frame and may be foam or any other suitable material.

[0038] The seat bottom **34** can include a frame and a covering. The frame may include tubes, beams, etc. Specifically, the frame may include a pair of frame members, e.g., elongated along the longitudinal axis **A2**. The frame members are spaced from each other along the lateral axis **A1**. The frame may include cross-members extending between the frame members. The cross-members may be elongated along the lateral axis **A1**. The frame can include a seat pan. The seat pan may be generally planar and extend from one of the frame members to the other of the frame members, e.g., along the lateral axis **A1**. The seat pan may be fixed to the frame members. The frame, including the frame members, the cross-members, and the seat pan may be of any suitable plastic material, e.g., carbon fiber reinforced plastic (CFRP), glass fiber-reinforced semi-finished thermoplastic composite (organosheet), etc. As another example, some or all components of the frame may be formed of a suitable metal, e.g., steel, aluminum, etc. The covering may include upholstery and padding. The upholstery may be cloth, leather, faux leather, or any other suitable material. The upholstery may be stitched in panels around the frame. The padding may be between the upholstery and the frame. The padding may be foam or any other suitable material. The seatback **32** is supported by the seat bottom **34** at a rear of the seat bottom **34**. The seat bottom **34** extends forward from the seatback **32**. The seatback **32** is movable relative to the seat bottom **34**. For example, the seatback **32** may pivot about an axis that is parallel to the longitudinal axis **A2**.

[0039] The middle console **20** of the vehicle **10** is adjacent, i.e., next to, the first seat **12** and the second seat **16**. The middle console **20** is disposed between the first seat **12** and the second seat **16**, e.g., along the lateral axis **A1**. The middle console **20** may be on a longitudinally extending midline of the vehicle **10**. The middle console **20** may be in the middle of two adjacent seats **12**, **16**, e.g., the first seat **12** and the second seat **16**. The middle console **20** may be elongated along the longitudinal axis **A2**. The middle console **20** may include an armrest, cupholders, etc.

[0040] The vehicle **10** includes airbags **14**, **18**, **36**, **38**, e.g., the first airbag **14**, the second airbag **18**, a third airbag **36**, and a fourth airbag **38**, for controlling kinematics of one or more occupants of the vehicle **10** during certain impacts to the vehicle **10**. Each of the airbags **14**, **18**, **36**, **38** is inflatable from an uninflated position, shown in FIG. 1, to an inflated position, shown in FIG. 2. The airbags **14**, **18**, **36**, **38** may each include panels of a woven polymer, or any other suitable material, that define inflation chambers that can receive inflation medium to inflate the airbags **14**, **18**, **36**, **38** to the inflated positions. As one example, the airbags **14**, **18**, **36**, **38** may be formed of woven nylon yarn, for example, nylon 6-6. Other examples include polyether ether ketone (PEEK), polyetherketoneketone (PEKK), polyester, etc. The woven polymer may include a coating, such as silicone, neoprene, urethane, etc. For example, the coating may be polyorgano siloxane.

[0041] The first airbag **14** is supported by the first seat **12**. The second airbag **18** is supported by the second seat **16**. The first airbag **14** and the second airbag **18** may be supported by the seatbacks **32** of the seats **12**, **16**. The first airbag **14** and the second airbag **18** may be fixed to frames of the seatbacks **32**, e.g., fixed to the pair of upright frame members via fasteners or other suitable structure. The first airbag **14** and the second airbag **18** may be supported via airbag housings, e.g., fixed to the frames of the seatbacks **32**. The first airbag **14** and the second airbag **18** in the uninflated positions may be internal of the first seat **12** and the second seat **16**, e.g., under the coverings of the seatbacks **32**.

[0042] The first airbag **14** and the second airbag **18** in the inflated positions may generally control kinematics of an occupant along the lateral axis **A1**. The first airbag **14** and the second airbag **18** in

the inflated positions extend forward from the seatbacks **32** of the first seat **12** and the second seat **16**. For example, the first airbag **14** in the inflated position may extend from the seatback **32** of the first seat **12** along the longitudinal axis **A2** between the seat bottom **34** of the first seat **12** and the middle console **20** along the lateral axis **A1**. The second airbag **18** in the inflated position may extend forward from the seatback **32** of the second seat **16** along the longitudinal axis **A2** and between the seat bottom **34** of the second seat **16** and the middle console **20** along the lateral axis **A1**. The first airbag **14** and the second airbag **18** in the inflated positions may be longer along the longitudinal axis **A2** than along the lateral axis **A1**.

[0043] The third airbag **36** and the fourth airbag **38** are supported by the middle console **20**. For example, the first airbag **14** and the second airbag **18** may be fixed within the middle console **20**, e.g., via fastener or other suitable structure. The third airbag **36** and the fourth airbag **38** may be supported via airbag housings, e.g., fixed to the middle console **20**. The third airbag **36** and the fourth airbag **38** in the uninflated positions may be internal of the middle console **20**.

[0044] The third airbag **36** and the fourth airbag **38** in the inflated positions may generally control kinematics of an occupant along the longitudinal axis **A2**. The third airbag **36** and the fourth airbag **38** in the inflated positions extend laterally outboard from the middle console **20** in front of the first seat **12** and the second seat **16**. For example, the third airbag **36** may extend from the middle console **20** along the lateral axis **A1** in front of the first seat **12**. The fourth airbag **38** may extend from the middle console **20** along the lateral axis **A1** in front of the second seat **16**. The first airbag **14** and the second airbag **18** in the inflated positions may be longer along the lateral axis **A1** than along the longitudinal axis **A2**.

[0045] The vehicle **10** may include additional airbags **14**, **18**, **36**, **38** (not shown). For example, the vehicle **10** may include one or more airbags **14**, **18**, **36**, **38** supported by other seats **12**, **16**, by an instrument panel of the vehicle **10**, by a steering wheel of the vehicle **10**, by a pillar of the body of the vehicle **10**, and/or any other suitable structure of the vehicle **10**.

[0046] The inflator **30**, shown in FIGS. **3-5**, provides inflation medium, e.g., to inflate the first airbag **14**, the second airbag **18**, the third airbag **36**, and/or the fourth airbag **38** from the uninflated positions to the inflated positions. The inflator **30** may be, for example, a pyrotechnic inflator that uses a chemical reaction to drive inflation medium to the first airbag **14** and/or the second airbag **18**. The inflator **30** may be of any suitable type, for example, a cold-gas inflator.

[0047] The inflator **30** is supported by the middle console **20**. For example, the inflator **30** may be fixed within the middle console **20**, e.g., via fastener or other suitable structure. The inflator **30** may be fixed to another component supported by the middle console **20**, e.g., to the manifold **22**.

[0048] The inflator **30** is operatively coupled to the manifold **22** to provide fluid to the chamber **24** of the manifold **22**. For example, the inflator **30** may be fixed to the manifold **22** and provide inflation medium directly to the chamber **24**. As another example, the inflator **30** may be connected to the manifold **22** via a fill tube **40** or other suitable structure that directs inflation medium from the inflator **30** to the chamber **24** of the manifold **22**.

[0049] The manifold **22**, shown in FIGS. **1-4**, is included to receive inflation medium from the inflator **30** and to provide the inflation medium to the first airbag **14**, the second airbag **18**, the third airbag **36**, and/or the fourth airbag **38**. The manifold **22** is supported by the middle console **20**. For example, the manifold **22** may be fixed within the middle console **20**, e.g., via fastener or other suitable structure. The manifold **22** may be metal, a fiber laminate, or any other suitable material.

[0050] The manifold **22** defines the chamber **24** to receive and provide the inflation medium. The manifold **22** encloses the chamber **24**. The chamber **24** may store inflation medium from the inflator **30** under pressure. The chamber **24** may be of sufficient volume such that generally constant pressure can be provided to the first airbag **14**, the second airbag **18**, the third airbag **36**, and/or the fourth airbag **38**. The manifold **22** may be a pressure vessel that stores a sufficient volume of fluid at sufficient pressure to fully inflate the first airbag **14**, the second airbag **18**, the third airbag **36**, and/or the fourth airbag **38** to the inflated positions. The volume of the chamber **24**

may be based on a sum of volumes of the airbags **14, 18, 36, 38** in the inflated positions and an amount of pressure that is sufficient to cause inflation of the airbags **14, 18, 36, 38** within a certain amount of time. The chamber **24** defines a cross-sectional area that is greater than a cross-sectional area defined by the first fluid outlet **26**. In other words, the cross-sectional area of the first fluid outlet **26** is less than the cross-sectional area of the chamber **24**. The cross-section area of the chamber **24** extends perpendicular to a flow direction of fluid through the chamber **24**.

[0051] The manifold **22** includes a plurality of fluid outlets **26, 28, 42, 44, 46**, e.g., the first fluid outlet **26**, the second fluid outlet **28**, a third fluid outlet **42**, a fourth fluid outlet **44**, and an exhaust fluid outlet **46**, for directing fluid flow out of the chamber **24**. The fluid outlets **26, 28, 42, 44, 46** provide passages that direct the flow of fluid. The fluid outlets **26, 28, 42, 44, 46** may be open to the chamber **24** on one end and open to external of the chamber **24** on an opposite end. The first fluid outlet **26** is operatively coupled to the first airbag **14** to provide fluid from the chamber **24** to the first airbag **14**. The second fluid outlet **28** is operatively coupled to the second airbag **18** to provide fluid from the chamber **24** to the second airbag **18**. The third fluid outlet **42** may be operatively coupled to the third airbag **36** to provide fluid from the chamber **24** to the third airbag **36**. The fourth fluid outlet **44** may be operatively coupled to the fourth airbag **38** to provide fluid from the chamber **24** to the fourth airbag **38**. For example, fill tube **40** may connect the first fluid outlet **26** to the first airbag **14**, the second fluid outlet **28** to the second airbag **18**, etc. The exhaust fluid outlet **46** enables fluid in the chamber **24** to be depressurized, e.g., the pressurized fluid may be exhausted from the chamber **24** without such fluid being directed to any airbag. For example, one end of the exhaust fluid outlet **46** may be open to the chamber **24** and an opposite end may be open to external of the chamber **24** and not operatively coupled to any airbag. The exhaust fluid outlet **46** may exhaust fluid, e.g., to the passenger cabin.

[0052] With reference to FIGS. 3-5, the vehicle **10** may include one or more valves **48, 50, 52, 54, 56**, e.g., a first valve **48**, a second valve **50**, a third valve **52**, a fourth valve **54**, and an exhaust valve **56**, for selectively permitting and inhibiting fluid flow through the fluid outlets **26, 28, 42, 44, 46**. Each valve **48, 50, 52, 54, 56** is movable to a closed position that inhibits fluid flow, e.g., as shown in FIG. 3. Each valve **48, 50, 52, 54, 56** is movable to an open position that permits fluid flow, e.g., as shown in FIG. 4. The valves **48, 50, 52, 54, 56** may be, for example, ball valves, butterfly valves, gate valves, or other suitable structure that selectively permits and inhibits fluid flow including in some examples those that are conventionally known. The valves **48, 50, 52, 54, 56** may move from the closed position to the open position, or vice versa, in response to a command from a computer **58**. Each valve **48, 50, 52, 54, 56** may include an actuator, an electromagnetic solenoid, a motor, and/or other suitable structure, including in some examples those that are conventionally known, for moving the respective valve from the closed position to the open position, or vice versa, in response to a command from the computer **58**.

[0053] The first valve **48** may be connected at the first fluid outlet **26**. The first valve **48** at the open position may permit fluid flow through the first fluid outlet **26** from the chamber **24** to the fill tube **40** operatively coupling the first fluid outlet **26** to the first airbag **14**. The first valve **48** at the closed position may inhibit fluid flow through the first fluid outlet **26** from the chamber **24** to the fill tube **40** operatively coupling the first fluid outlet **26** to the first airbag **14**.

[0054] The second valve **50** may be connected at the second fluid outlet **28**. The second valve **50** at the open position may permit fluid flow through the second fluid outlet **28** from the chamber **24** to the fill tube **40** operatively coupling the second fluid outlet **28** to the second airbag **18**. The second valve **50** at the closed position may inhibit fluid flow through the second fluid outlet **28** from the chamber **24** to the fill tube **40** operatively coupling the second fluid outlet **28** to the second airbag **18**.

[0055] The third valve **52** may be connected at the third fluid outlet **42**. The third valve **52** at the open position may permit fluid flow through the third fluid outlet **42** from the chamber **24** to the fill tube **40** operatively coupling the third fluid outlet **42** to the third airbag **36**. The third valve **52** at the

closed position may inhibit fluid flow through the third fluid outlet **42** from the chamber **24** to the fill tube **40** operatively coupling the third fluid outlet **42** to the third airbag **36**.

[0056] The fourth valve **54** may be connected at the fourth fluid outlet **44**. The fourth valve **54** at the open position may permit fluid flow through the fourth fluid outlet **44** from the chamber **24** to the fill tube **40** operatively coupling the fourth fluid outlet **44** to the fourth airbag **38**. The fourth valve **54** at the closed position may inhibit fluid flow through the fourth fluid outlet **44** from the chamber **24** to the fill tube **40** operatively coupling the fourth fluid outlet **44** to the second airbag **18**.

[0057] The exhaust valve **56** may be connected at the exhaust fluid outlet **46**. The exhaust valve **56** at the open position may permit fluid flow out of the chamber **24** through the exhaust fluid outlet **46**. The exhaust valve **56** at the closed position may inhibit fluid flow out of the chamber **24** through the exhaust fluid outlet **46**.

[0058] With reference to FIG. 5, the vehicle **10** may include at least one impact sensor **60**. The impact sensor **60** is configured to detect certain vehicle impacts. A “certain vehicle impact” is an impact of the type and/or magnitude for which inflation of the one or more of the airbags **14**, **18**, **36**, **38** is designed i.e., “certain” indicates the type and/or magnitude of the impact. The type and/or magnitude of such “certain vehicle impacts” may be pre-stored in the computer **58**, e.g., a restraints control module and/or a body control module. The impact sensor **60** may be configured to detect certain impact contemporaneously with the certain impact and/or the impact sensor **60** may be configured to sense certain impacts prior to impact, i.e., pre-impact sensing. The impact sensor **60** may be of any suitable type, for example, post contact sensors such as accelerometers, pressure sensors, and contact switches; and pre-impact sensors **60** such as radar, LIDAR, and vision sensing systems. The vision systems may include one or more cameras, CCD image sensors, CMOS image sensors, etc. The impact sensors **60** may be located at numerous points in or on the vehicle **10**.

[0059] The vehicle **10** may include a communication network **62**. The communication network **62** includes hardware, such as a communication bus, for facilitating communication among components of the vehicle **10**, e.g., the inflator **30**, the valves **48**, **50**, **52**, **54**, **56**, the impact sensor **60**, the computer **58**, etc. The communication network **62** may facilitate wired or wireless communication among the components in accordance with a number of communication protocols such as controller area network (CAN), Ethernet, WiFi, Local Interconnect Network (LIN), and/or other wired or wireless mechanisms. Alternatively or additionally, in cases where the computer **58** comprises a plurality of devices, the communication network **62** may be used for communications between devices represented as the computer **58** in this disclosure.

[0060] The computer **58** may be, for example, a restraints control module of the vehicle **10**. The computer **58** includes a processor, a memory, etc. The memory of the computer **58** may include memory for storing programming instructions executable by the processor as well as for electronically storing data and/or databases. The computer **58** may be a microprocessor-based computer implemented via circuits, chips, or other electronic components. For example, the computer **58** can be a generic computer with a processor and memory as described above and/or may include an electronic control unit (ECU) or controller for a specific function or set of functions, and/or a dedicated electronic circuit including an ASIC that is manufactured for a particular operation, e.g., an ASIC for processing sensor data and/or communicating the sensor data. In another example, computer **58** may include an FPGA (Field-Programmable Gate Array) which is an integrated circuit manufactured to be configurable by a user. Typically, a hardware description language such as VHDL (Very High-Speed Integrated Circuit Hardware Description Language) is used in electronic design automation to describe digital and mixed-signal systems such as FPGA and ASIC. For example, an ASIC is manufactured on VHDL programming provided pre-manufacturing, whereas logical components inside an FPGA may be configured based on VHDL programming, e.g., stored in a memory electrically connected to the FPGA circuit. In some examples, a combination of processor(s), ASIC(s), and/or FPGA circuits may be included in the

computer **58**. The memory can be of any type, e.g., hard disk drives, solid state drives, servers, or any volatile or non-volatile media. The memory can store the collected data sent from the sensors. [0061] The computer **58** is programmed to, i.e., the memory stores instructions executable by the processor to, detect one or more certain impacts to the vehicle **10**. The computer **58** may detect one or more certain impacts based on information received, e.g., via the communication network **62**, from the impact sensors **60**. The information received from the impact sensors **60** may indicate, e.g., a location of impact, angle of impact, amount of force of impact, acceleration of the vehicle **10**, etc.

[0062] The computer **58** may be programmed to command actuation of the inflator **30** to provide inflation medium to the manifold **22**. The computer **58** may command actuation of the inflator **30** by transmitting a command to the inflator **30**, e.g., via the communication network **62**. The command may include, for example, an electrical pulse that triggers expansion of the inflation medium.

[0063] The computer **58** is programmed to command one or more of the valves **48, 50, 52, 54, 56** from the closed position to the open position, and vice versa. The computer **58** may command one or more of the valves **48, 50, 52, 54, 56** to the open position or the closed position by transmitting one or more commands specifying the open position or the closed position to one or more of the valves **48, 50, 52, 54, 56**. For example, the computer **58** may command the first valve **48** from the closed position to the open position by transmitting a command to the first valve **48**, the command specifying the open position. As another example, the computer **58** may command the exhaust valve **56** from the open position to the closed position by transmitting a command to the exhaust valve **56**, the command specifying the closed position.

[0064] The computer **58** is programmed to actuate the inflator **30** and the valves **48, 50, 52, 54, 56** to control inflation of the first airbag **14** the second airbag **18**, the third airbag **36**, and/or the fourth airbag **38**, e.g., in response to detecting one or more certain impacts to the vehicle **10**. For example, during normal operation of the vehicle **10**, i.e., prior to the computer **58** detecting certain impact to the vehicle **10**, the valves **48, 50, 52, 54, 56** may all be at the closed positions, e.g., as shown in FIG. **3**. In response to detecting a certain impact to the vehicle **10**, the computer **58** commands actuation of the inflator **30** to provide inflation medium to the manifold **22**. Also in response to detecting the certain impact to the vehicle **10**, the computer **58** may command at least one of the first valve **48**, the second valve **50**, the third valve **52**, and or the fourth valve **54** from the closed position to the open position to permit inflation medium to flow from the manifold **22** to the first airbag **14** the second airbag **18**, the third airbag **36**, and/or the fourth airbag **38**.

[0065] The computer **58** may be programmed to inflate additional of the first airbag **14** the second airbag **18**, the third airbag **36**, and/or the fourth airbag **38** in response to detecting another certain impact to the vehicle **10** after actuating the inflator **30**. For example, in response to detecting the another certain impact to the vehicle **10**, the computer **58** may command another of the first valve **48** the second valve **50**, the third valve **52**, and/or or the fourth valve **54** from the closed position to the open position to permit inflation medium to flow from the manifold **22** to other of the first airbag **14**, the second airbag **18**, the third airbag **36**, and/or the fourth airbag **38** that have not been previously inflated.

[0066] The computer **58** may select which of the first valve **48**, the second valve **50**, the third valve **52**, and/or or the fourth valve **54** to command from the closed position to the open position based on the certain detected impact, e.g., depending on whether the certain impact is a frontal impact FI (shown in FIG. **6**), a side impact SI (shown in FIG. **7**), an offset impact OI to the first side (shown in FIG. **8**), or an offset impact to the second side (not shown). For example, the computer **58** may store a look-up table or the like associating opening various of the valves **48, 50, 52, 54** with the different certain impacts. A sample look-up is provided below in Table 1.

TABLE-US-00001 TABLE 1 Offset Offset Side Frontal Impact - Impact - Impact Impact First Side
Second Side First valve 48 Open na na Open Second valve 50 Open na Open na Third valve 52 na

Open Open na Fourth valve 54 na Open na Open

[0067] The computer **58** may be programmed to command the first valve **48**, the second valve **50**, the third valve **52**, and/or the fourth valve **54** from the open positions to the closed positions after a threshold amount of time has lapsed from commanding such valves **48**, **50**, **52**, **54** from the closed position to the open position, e.g., to control a inflation pressure of the respective airbag **14**, **18**, **36**, **38** in the inflated position. A longer amount of time between opening and then closing the valves **48**, **50**, **52**, **54** permits more inflation medium to flow into the respective airbag **14**, **18**, **36**, **38**, thereby providing a relatively higher inflation pressure to the respective airbag **14**, **18**, **36**, **38**. A shorter amount of time between opening and then closing the valves **48**, **50**, **52**, **54** permits less inflation medium to flow into the respective airbag **14**, **18**, **36**, **38**, thereby providing a lower inflation pressure to the respective airbag **14**, **18**, **36**, **38**. The computer **58** may determine the threshold amount of time based on a detected size of an occupant of the first seat **12** or the second seat **16**, e.g., to provide a higher airbag pressure for a larger occupant and a lower airbag pressure for a smaller occupant. The threshold amount of time may be empirically determined and/or calculated, e.g., based on crash test data, computer modeling, etc. The threshold amount of time is greater than zero and may be, e.g., 20 to 60 milliseconds.

[0068] The computer **58** may be programmed to command the exhaust valve **56** from the closed position to the open position after a threshold amount of time has lapsed from detecting the certain impact, e.g., to release any remaining pressurized inflation medium from the manifold **22**. The threshold amount of time may be sufficient to allow the manifold **22** to retain the pressurized inflation medium to inflate one or more additional airbags **14**, **18**, **36**, **38** should another certain impact be detected after detection of the initial certain impact. The threshold amount of time may be, for example, at least two seconds.

[0069] In general, the computing systems and/or devices described may employ any of a number of computer operating systems, including, but by no means limited to, versions and/or varieties of the Ford Sync® application, AppLink/Smart Device Link middleware, the Microsoft Automotive® operating system, the Microsoft Windows® operating system, the Unix operating system (e.g., the Solaris® operating system distributed by Oracle Corporation of Redwood Shores, California), the AIX UNIX operating system distributed by International Business Machines of Armonk, New York, the Linux operating system, the Mac OSX and iOS operating systems distributed by Apple Inc. of Cupertino, California, the BlackBerry OS distributed by Blackberry, Ltd. of Waterloo, Canada, and the Android operating system developed by Google, Inc. and the Open Handset Alliance, or the QNX® CAR Platform for Infotainment offered by QNX Software Systems. Examples of computing devices include, without limitation, an on-board vehicle computer, a computer workstation, a server, a desktop, notebook, laptop, or handheld computer, or some other computing system and/or device.

[0070] Computing devices generally include computer-executable instructions, where the instructions may be executable by one or more computing devices such as those listed above. Computer executable instructions may be compiled or interpreted from computer programs created using a variety of programming languages and/or technologies, including, without limitation, and either alone or in combination, Java™, C, C++, Matlab, Simulink, Stateflow, Visual Basic, Java Script, Python, Perl, HTML, etc. Some of these applications may be compiled and executed on a virtual machine, such as the Java Virtual Machine, the Dalvik virtual machine, or the like. In general, a processor (e.g., a microprocessor) receives instructions, e.g., from a memory, a computer readable medium, etc., and executes these instructions, thereby performing one or more processes, including one or more of the processes described herein. Such instructions and other data may be stored and transmitted using a variety of computer readable media. A file in a computing device is generally a collection of data stored on a computer readable medium, such as a storage medium, a random access memory, etc.

[0071] A computer-readable medium (also referred to as a processor-readable medium) includes

any non-transitory (e.g., tangible) medium that participates in providing data (e.g., instructions) that may be read by a computer (e.g., by a processor of a computer). Such a medium may take many forms, including, but not limited to, non-volatile media and volatile media. Instructions may be transmitted by one or more transmission media, including fiber optics, wires, wireless communication, including the internals that comprise a system bus coupled to a processor of a computer. Common forms of computer-readable media include, for example, RAM, a PROM, an EPROM, a FLASH-EEPROM, any other memory chip or cartridge, or any other medium from which a computer can read.

[0072] Databases, data repositories or other data stores described herein may include various kinds of mechanisms for storing, accessing, and retrieving various kinds of data, including a hierarchical database, a set of files in a file system, an application database in a proprietary format, a relational database management system (RDBMS), a nonrelational database (NoSQL), a graph database (GDB), etc. Each such data store is generally included within a computing device employing a computer operating system such as one of those mentioned above, and are accessed via a network in any one or more of a variety of manners. A file system may be accessible from a computer operating system, and may include files stored in various formats. An RDBMS generally employs the Structured Query Language (SQL) in addition to a language for creating, storing, editing, and executing stored procedures, such as the PL/SQL language mentioned above.

[0073] In some examples, system elements may be implemented as computer-readable instructions (e.g., software) on one or more computing devices (e.g., servers, personal computers, etc.), stored on computer readable media associated therewith (e.g., disks, memories, etc.). A computer program product may comprise such instructions stored on computer readable media for carrying out the functions described herein.

[0074] With regard to the media, processes, systems, methods, heuristics, etc. described herein, it should be understood that, although the steps of such processes, etc. have been described as occurring according to a certain ordered sequence, such processes could be practiced with the described steps performed in an order other than the order described herein. It further should be understood that certain steps could be performed simultaneously, that other steps could be added, or that certain steps described herein could be omitted. Operations, systems, and methods described herein should always be implemented and/or performed in accordance with an applicable owner's/user's manual and/or safety guidelines.

[0075] The numerical adjectives “first,” “second,” etc., are used herein merely as identifiers and do not signify order or importance. Use of “in response to,” “based on,” and “upon” herein indicates a causal relationship, not merely a temporal relationship.

[0076] The disclosure has been described in an illustrative manner, and it is to be understood that the terminology which has been used is intended to be in the nature of words of description rather than of limitation. Many modifications and variations of the present disclosure are possible in light of the above teachings, and the disclosure may be practiced otherwise than as specifically described.

Claims

1. A vehicle, comprising: a first seat; a first airbag supported by the first seat; a second seat; a second airbag supported by the second seat; a middle console disposed between the first seat and the second seat; a manifold supported by the middle console, the manifold defining a chamber and including a first fluid outlet operatively coupled to the first airbag to provide fluid from the chamber to the first airbag and a second fluid outlet operatively coupled to the second airbag to provide fluid from the chamber to the second airbag; and an inflator supported by the middle console, the inflator operatively coupled to the manifold to provide fluid to the chamber.
2. The vehicle of claim 1, wherein the manifold includes an exhaust fluid outlet.

3. The vehicle of claim 2, further comprising an exhaust valve movable to an open position permitting fluid flow through the exhaust fluid outlet and to a closed position inhibiting fluid flow through the exhaust fluid outlet.
4. The vehicle of claim 3, further comprising a first valve movable to an open position permitting fluid flow through the first fluid outlet and to a closed position inhibiting fluid flow through the first fluid outlet, and a second valve movable to an open position permitting fluid flow through the second fluid outlet and to a closed position inhibiting fluid flow through the second fluid outlet.
5. The vehicle of claim 4, further comprising a computer having a processor and memory storing instructions executable by the processor to, in response to detecting a certain impact to the vehicle, command at least one of the first valve or the second valve from the closed position to the open position and command actuation of the inflator.
6. The vehicle of claim 5, wherein the instructions include instructions to, after a threshold amount of time has lapsed from commanding the at least one of the first valve or the second valve from the closed position to the open position, command the at least one of the first valve or the second valve from the open positions to the closed positions.
7. The vehicle of claim 6, wherein the threshold amount of time is 20 to 60 milliseconds.
8. The vehicle of claim 6, wherein the instructions include instructions to command the exhaust valve from the closed position to the open position after a second threshold amount of time has lapsed from detecting the certain impact.
9. The vehicle of claim 8, wherein the second threshold amount of time is at least two seconds.
10. The vehicle of claim 1, further comprising a third airbag supported by the middle console and a fourth airbag supported by the middle console, and wherein the manifold includes a third fluid outlet operatively coupled to the third airbag to provide fluid from the chamber to the third airbag and a fourth fluid outlet operatively coupled to the fourth airbag to provide fluid from the chamber to the fourth airbag.
11. The vehicle of claim 10, wherein the first seat and the second seat each include a seatback, wherein the first airbag and the second airbag are inflatable to inflated positions that extend forward from the seatbacks of the first seat and the second seat, and wherein the third airbag and the fourth airbag are inflatable to inflated positions that extend laterally outboard from the middle console in front of the first seat and the second seat.
12. The vehicle of claim 10, wherein the manifold includes an exhaust fluid outlet.
13. The vehicle of claim 12, further comprising an exhaust valve moveable to an open position permitting fluid flow through the exhaust fluid outlet and to a closed position inhibiting fluid flow through the exhaust fluid outlet.
14. The vehicle of claim 13, further comprising a first valve movable to an open position permitting fluid flow through the first fluid outlet and to a closed position inhibiting fluid flow through the first fluid outlet, a second valve movable to an open position permitting fluid flow through the second fluid outlet and to a closed position inhibiting fluid flow through the second fluid outlet, a third valve movable to an open position permitting fluid flow through the third fluid outlet and to a closed position inhibiting fluid flow through the third fluid outlet, and a fourth valve movable to an open position permitting fluid flow through the fourth fluid outlet and to a closed position inhibiting fluid flow through the fourth fluid outlet.
15. The vehicle of claim 14, further comprising a computer having a processor and memory storing instructions executable by the processor to, in response to detecting a certain impact to the vehicle, command at least one of the first valve, the second valve, the third valve, or the fourth valve from a closed position to an open position and command actuation of the inflator.
16. The vehicle of claim 15, wherein the instructions include instructions to, in response to detecting another certain impact to the vehicle, command another of the at least one of the first valve, the second valve, the third valve, or the fourth valve from the closed position to the open position.

17. The vehicle of claim 16, wherein instructions include instructions to command the exhaust valve from the closed position to the open position after a threshold amount of time has lapsed from detecting the certain impact.

18. The vehicle of claim 17, wherein the threshold amount of time is at least two seconds.

19. The vehicle of claim 1, wherein the chamber defines a first cross-sectional area, and the first fluid outlet defines a second cross-sectional area that is less than the first cross-sectional area.
